Lucas Barton

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Mechanical engineer with 7+ years experience in product development and precision hardware design, including mass spectrometry and laser systems, as well as thermal and airflow management to IP54 standards. Now leveraging that expertise to develop solutions in the HVAC industry.

Relevant Experience

908 Devices — Boston, MA
Mechanical Engineer — November 2021 - November 2024

- Mechanical lead on MX908 Beacon, an area monitoring device for aerosol and vapor threats; developed product from concept through beta deployment and production, earning a Gold Homeland Security Award
- Engineered thermal and airflow systems for compact, sealed enclosures to ensure stable operating temps and IP54 protection, improving field reliability under extreme conditions
- Coordinated pilot builds and managed field units, collecting performance data that directly informed design revisions, improving product integrity and serviceability
- Applied CAD, 3D-printing, and sheet-metal expertise to produce precise microfluidics and ruggedized housings, accelerating design-to-test cycles and enabling quicker deployment of new products

Teradiode - a Panasonic Company — Wilmington, MA
Mechanical Engineer — October 2019 - November 2021

- Ownership of opto-mechanical laser engine system including design, prototyping, and floor builds
- Maintained and optimized 1000+ part assemblies with SolidWorks PDM, leveraging FEA and GD&T to improve system reliability, manufacturability (DFM), and overall performance efficiency
- Built and engineered custom fixtures for precise testing on optical parts in an ISO 7 clean room environment
- Designed complex metal, sheet metal, and electric parts and assemblies with high tolerances resulting in process and manufacturing improvements

Tactus — Boston, MA
Co-Founder and Chief Product Officer — August 2019 - May 2023

- Developed a wearable device that enables Deaf and Hard of Hearing to feel music via tactile sensation
- MassChallenge startup alumn responsible for technical design, layout and manufacturing processes
- Lead inventor on U.S. Patent No. 11,140,472 utilizing signal processing and strategically placed transducers
- Collaborated with customer early in design cycle, informing functional and comfort-driven requirements

Fortify — Boston, MA
Mechanical Engineering Co-op — July - December 2018

- Evaluated and tested Fortify's 3D printing systems to drive solutions in a small start-up environment
- Designed injection molds and components for in house 3D printers with Creo ProE and FEA software
- Developed sensor firmware and analyzed collected data to optimize system performance, providing actionable insights for real-world operational improvements
- Fabricated sheet metal assemblies, electromechanical systems, and prototypes through machining, soldering, tolerance analysis, and fluids analysis

Education

Northeastern University — Boston, MA
Bachelor of Science in Mechanical Engineering — May 2019

Honors: Dean's List, Presidential Global Scholarship, Dialogue of Civilizations Scholarship
Relevant Coursework: Heat Transfer, System Analysis & Control, Fluid Mechanics, FEA, Mechanics & Design, Dynamics & Vibrations, Alternative Energy, Biomechanics
Activities: Enabling Engineering, ASME, Jazz Ensemble, NUSound, Pep Band
Study Abroad: Alternative Energy Technology & Brazilian Culture — São Paulo, Brazil